



Constrained Auto-Regressive Decoding Constrains Generative Retrieval 约束自回归解码限制生成式检索



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Shiguang Wu
Shandong University
Qingdao, China
shiguang.wu@mail.sdu.edu.cn

Zhaochun Ren*
Leiden University
Leiden, The Netherlands
z.ren@liacs.leidenuniv.nl

Xin Xin
Shandong University
Qingdao, China
xinxin@sdu.edu.cn

Jiyuan Yang
Shandong University
Qingdao, China
jiyuan.yang@mail.sdu.edu.cn

Mengqi Zhang
Shandong University
Qingdao, China
mengqi.zhang@sdu.edu.cn

Zhumin Chen
Shandong University
Qingdao, China
chenzhumin@sdu.edu.cn

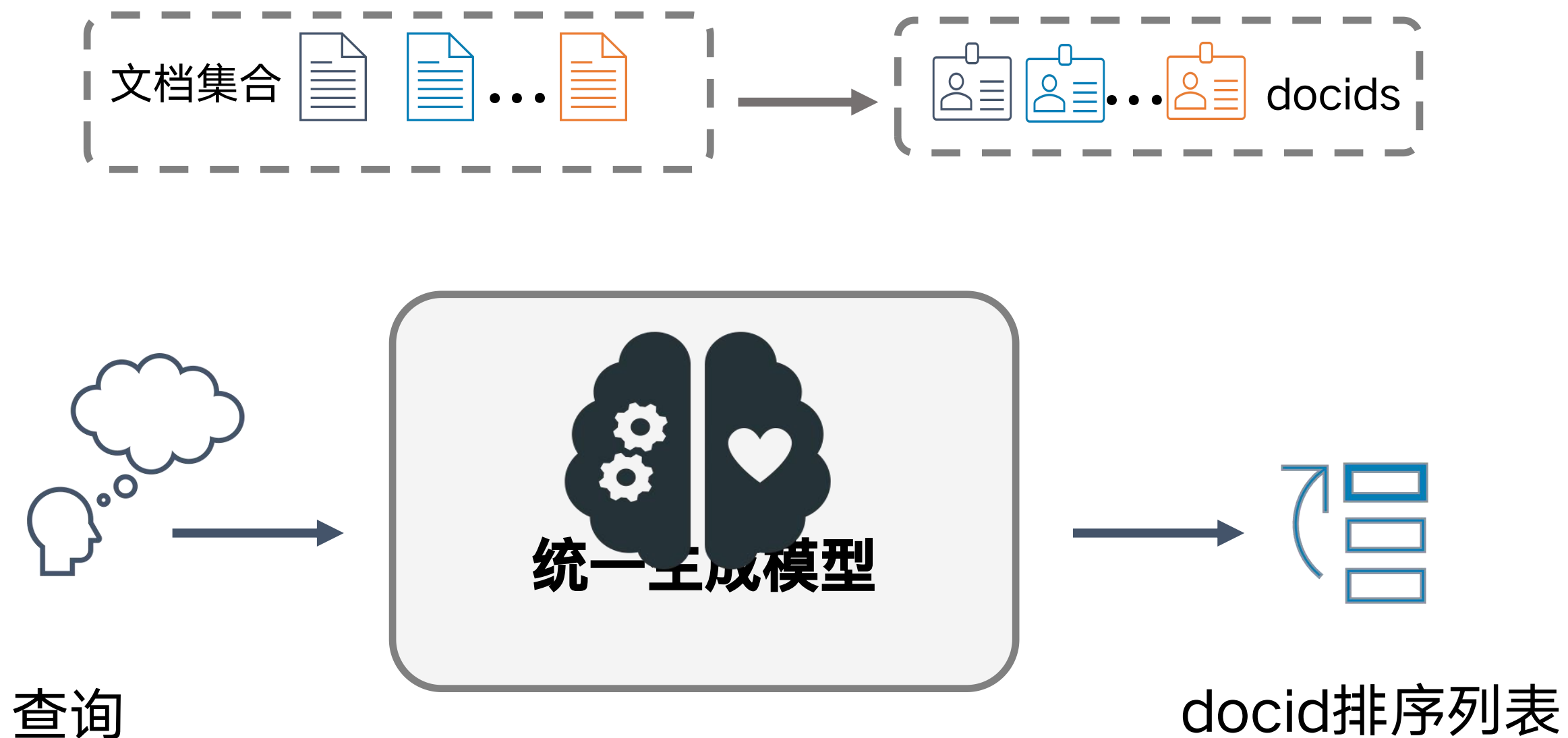
Maarten de Rijke
University of Amsterdam
Amsterdam, The Netherlands
m.derijke@uva.nl

Pengjie Ren*
Shandong University
Qingdao, China
jay.ren@outlook.com

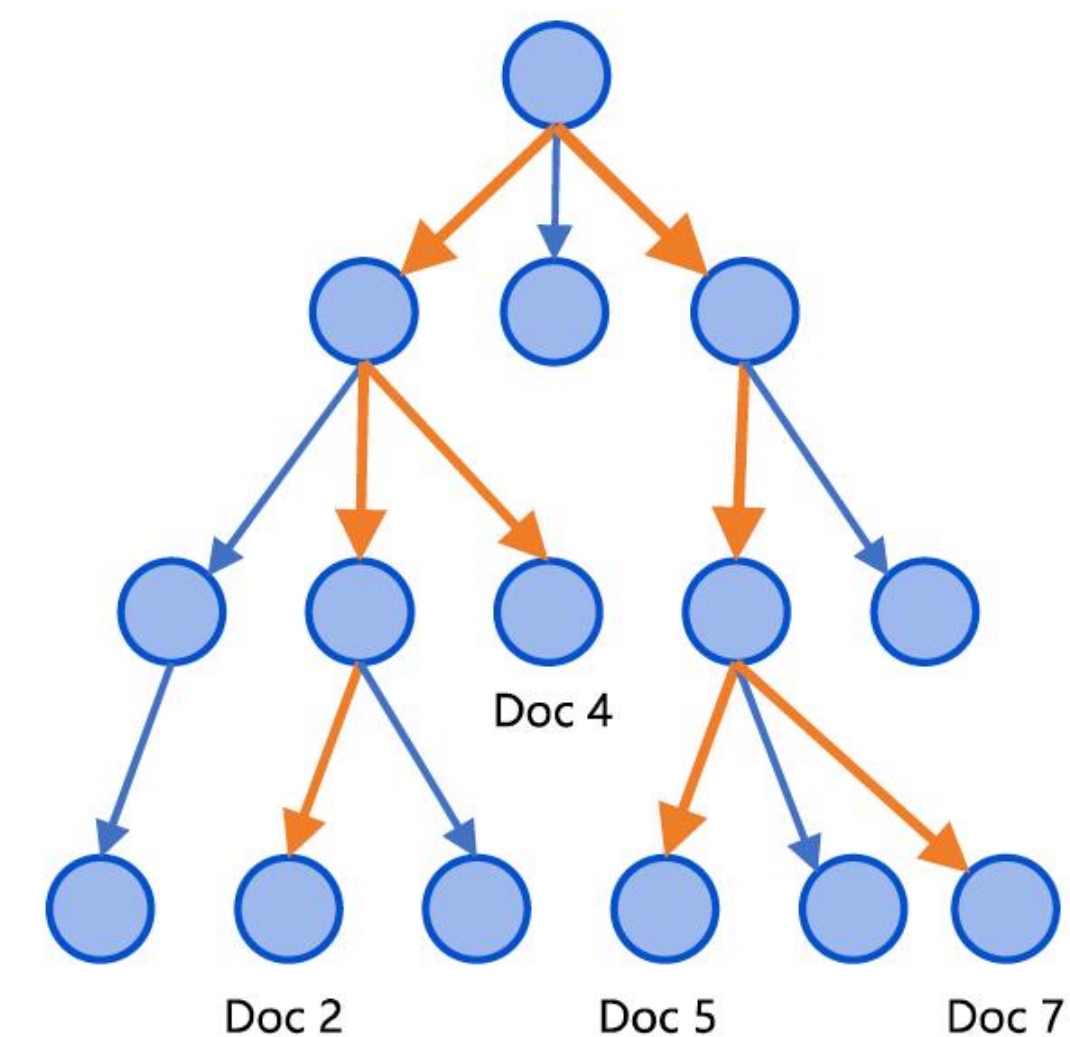
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» Background

Generative retrieval (GR)



Constrained beam search



- 将每个生成的docid限制在有效的预定义候选集中
- 定义一棵前缀树，结点是预定义候选集中的token

■ How the generation process affects the generalization of GR?

- Unlike dense retrieval, GR, as a retrieval paradigm itself, is much more concerned with the **generalization abilities to unseen corpora**.
- Recent advances on GR primarily focus on the problems arising from **training strategies**, and addressing them through various **learning techniques**.
- The fundamental challenges of generalization arising from **constrained auto-regressive decoding** still **remain unexplored and systematically understudied**.

■ Bayes-optimal generative retrieval

- Able to predict the ground-truth relevance distribution over $\mathcal{D}$ given any query.

Table 1: Glossary.

Symbol	Description
k, m	the vocabulary size and document length
$\mathcal{D}$	the complete corpus, i.e., $[k]^m$
$\mathcal{D}^c, C$	downstream corpus and constraints
f, f^c	Bayes-optimal and induced downstream GR
$\Pr(\cdot q)$	relevance distribution given query q
$d = (d_1, \dots, d_m)$	document in $\mathcal{D}$

Notation. Table 1 lists the main notation used in the paper. For an integer n , we denote the set $\{1, \dots, n\}$ by $[n]$. We use bold face to denote random variables. Tokens in a document d are integers from $[k]$, making $d \in [k]^m$, and k is the vocabulary size. We set the complete corpus $\mathcal{D} = [k]^m$. We denote the underlying relevance distribution over $\mathcal{D}$ given a query q as $\Pr(\cdot | q)$, where $d \sim \Pr(d | q) = \prod \Pr(d_i | d_{<i}, q)$. For simplicity, we focus on a single query q and omit it in some context. We use subscripts to indicate a sliced subset or operations at specific steps, i.e., $\square_i$, and $\square_{\geq i}$, etc. Particularly, we refer to $\mathcal{D}_{\geq i}$ or $\mathcal{D}_{\geq i}^c$ a branch with root d_i , under some prefix $d_{<i}$.

Downstream corpus and constraints. We construct the downstream corpus $\mathcal{D}^c$ by sampling. For simplicity, each document is sampled with an equal probability p . In practice, the sampling is agnostic to the future user queries, and thus independent with $\Pr(\cdot)$. We use C to be the event that d is in $\mathcal{D}^c$, and $\Pr(\cdot | C)$ is the distribution under the corpus $\mathcal{D}^c$. C_i means the i -th token of document d is valid with respect to the downstream corpus constraints, given some context-clear prefix tokens $d_{<i}$, i.e., d_i is valid if it appears in $\mathcal{D}^c$ for the prefix $d_{<i}$. We use “constraints”, and “downstream corpus” to represent the result of sampling interchangeably. The constrained generation process g_i^c is applied at the i -th step of f . It first zeros out the invalid tokens and then re-normalizes the remaining probabilities.

» Theoretical analysis

■ Constraints cause marginal distribution mismatch

- Only consider **the first generation step** without loss of generality. Other cases can be reduced to it by adjusting the document length m or vocabulary size k
- Target: Difference between **ground-truth** and **model-predicted step-wise** marginal distributions.

» Theoretical analysis

■ Constraints cause marginal distribution mismatch

- Model-predicted step-wise distribution

$$\Pr(\mathbf{d}_1 \mid q, C_1) = g^c [\Pr(\mathbf{d}_1 \mid q)] \quad (1)$$

$$\propto \mathbb{I}[d_1 \in \mathcal{D}_1^c] \sum_{\forall d_{>1} \in \mathcal{D}_{>1}} \Pr(d_1 d_{>1} \mid q), \quad (2)$$

- Marginal distribution

$$\Pr(\mathbf{d}_1 \mid q, C_1, \mathcal{C}_{>1}) = g^c [\Pr(\mathbf{d}_1 \mid q, \mathcal{C}_{>1})] \quad (3)$$

$$\propto \mathbb{I}[d_1 \in \mathcal{D}_1^c] \sum_{\forall d_{>1} \in \mathcal{D}_{>1}^c} \Pr(\mathbf{d}_1 d_{>1} \mid q), \quad (4)$$

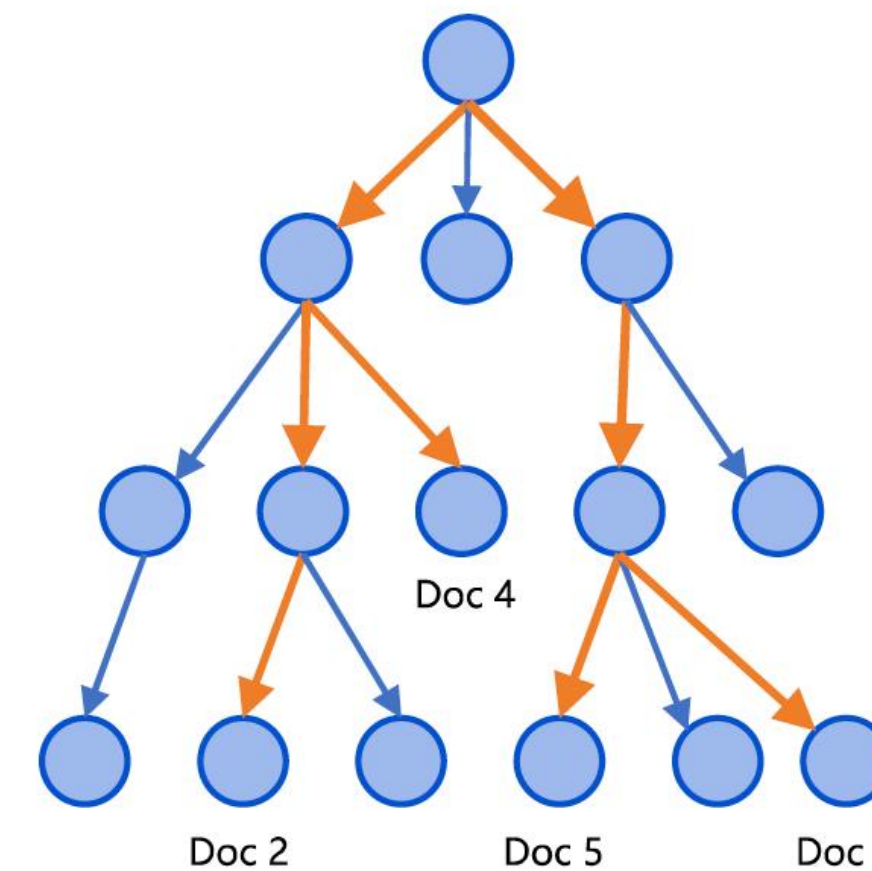
- Kullback–Leibler (KL) divergence

$$\text{KL} [\Pr(\cdot \mid C) \parallel \Pr(\cdot \mid C_1)] \quad (5)$$

$$= \mathbb{E}_{\mathbf{d}_1 \sim \Pr(\cdot \mid C)} \left[\log \frac{\Pr(\mathbf{d}_1 \mid C)}{\Pr(\mathbf{d}_1 \mid C_1)} \right] \quad (6)$$

$$= \mathbb{E} [\log \Pr(C_{>1} \mid \mathbf{d}_1)] - \log \Pr(C_{>1} \mid C_1). \quad (7)$$

- Root cause of distribution mismatch
Unawareness of future constraints.



- Key insight

We see that the KL divergence is the gap between **the average proportion of constraints locally** within each branch and the **global average constraints**.

The variation across branches may contribute the mismatch.

» Theoretical analysis

■ Constraints cause marginal distribution mismatch

$$\mathbb{E}_C[\Pr(d_1 \mid q, C)] \propto \mathbb{E}[\mathbb{I}[\mathbf{d} \in \mathcal{D}^c]] \Pr(d_1 \mid q) \propto \Pr(d_1 \mid q). \quad (8)$$

- Therefore, the downstream corpus will follow the same distribution as the complete corpus on average.
- However, the gap is not concentrated at zero with high probability in terms of KL divergence.
- Exceptional case: $\Pr(\mathbf{d} \mid q)$ is a uniform relevance distribution.

$$\text{KL} [\Pr(\cdot \mid C) \parallel \Pr(\cdot \mid C_1)] \gtrsim \frac{0.05}{k^{r-1}}. \quad (9)$$

- General case: $\Pr(\mathbf{d} \mid q)$ is a general relevance distribution.

$$\text{KL} [\Pr(\cdot \mid C) \parallel \Pr(\cdot \mid C_i)] \gtrsim \frac{0.05A}{p}. \quad (11)$$

Simpson diversity index.

$$A = \mathbb{E}_{d_1}^2 \left[\sqrt{\sum_d \Pr(d \mid d_1)^2} \right].$$

Summary: For the effect of constraints, we derive a lower bound of the error, in terms of the KL divergence between the ground-truth and the model-predicted step-wise marginal distributions. This error arises due to **the unawareness of future constraints** during generation and is shown to **depend on the average Simpson diversity index** of the relevance distribution.

» Theoretical analysis

■ The impact of beam search on recall

- Root cause: Non-relevant branches overtaking relevant ones.
- In summary, GR models relying on the sum of sub-tree values for ranking branches **struggle to achieve high recall performance while maintaining top-1 precision.**

» Concentration as a trade-off factor

- We find a common dependence on **the concentration of relevance distribution** from both components, i.e., the Simpson diversity index, and the thickness of the tail distribution.
- Concentration ↓ Width Diversity A ↓ Lower bound ↓
Have better alignment with the ground-truth constrained marginal distribution.
- Concentration Width ↓ thickness of the tail distribution false positive ↓
Improve the recall performance using beam search.

» Difficulty of concentration in Bayes-optimal GR

- **Aggregation:** Aggregating relevant documents within a narrow branch to ensure that its value significantly exceeds that of others throughout the generation process.
 - Aggregation introduces redundancy.
- **Amplification:** Amplifying the scores of relevant documents relative to non-relevant ones, so that the branch value serves as a reliable sketch for branch quality.
 - Amplification requires high-quality data.

» Synthetic experiments

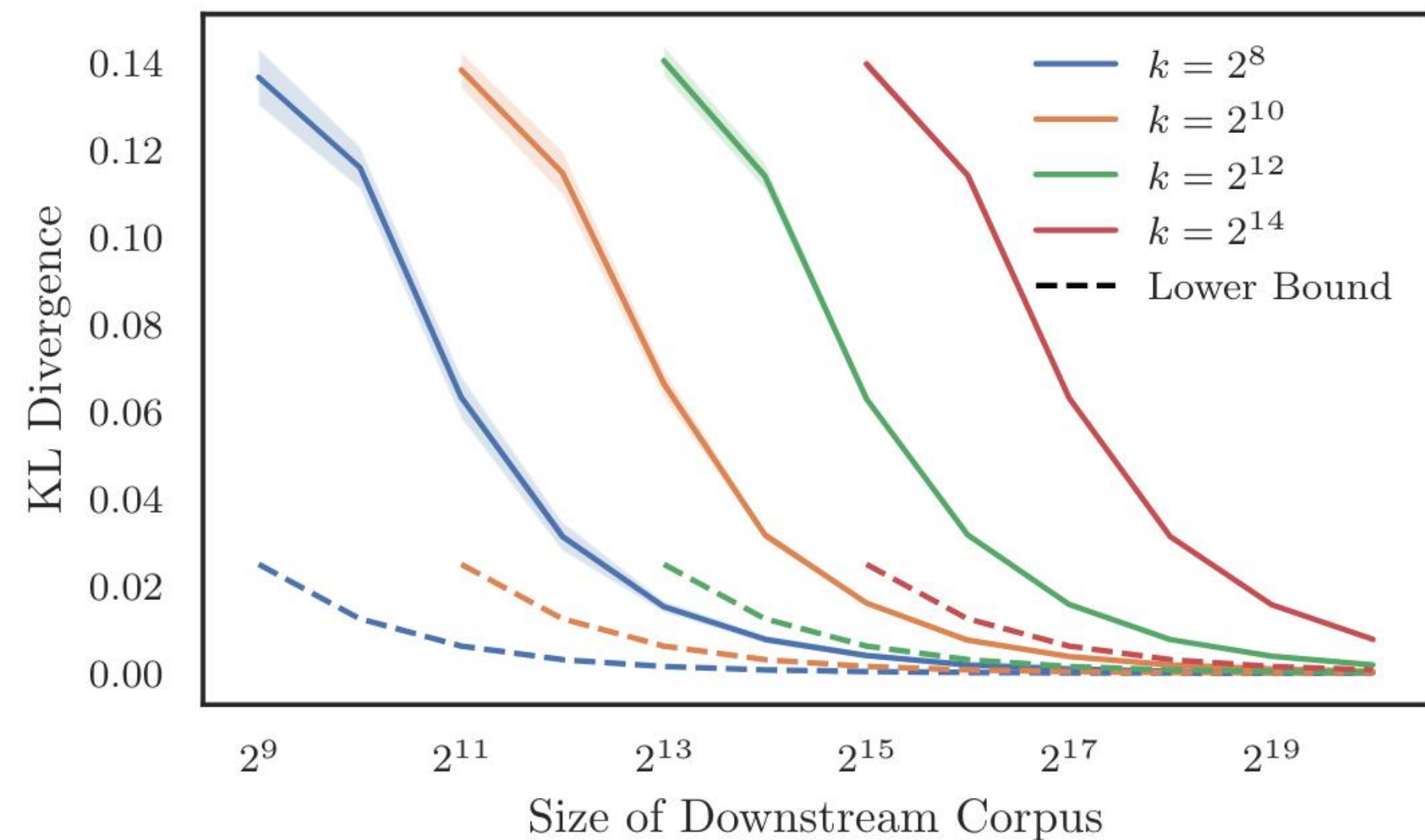


Figure 1: The KL divergence error in the first generation step on synthetic uniform relevance distribution data with uniformly sampled downstream corpus.

$$\text{KL} [\text{Pr}(\cdot | C) \parallel \text{Pr}(\cdot | C_1)] \gtrsim \frac{0.05}{k^{r-1}}. \quad (9)$$

$$k^{r-1} = k^{\log_k n - 1} = \frac{k^{\log_k n}}{k} = \frac{n}{k}$$

因此，公式 9 的下界可改写为： $\text{KL} \gtrsim \frac{0.05k}{n}$

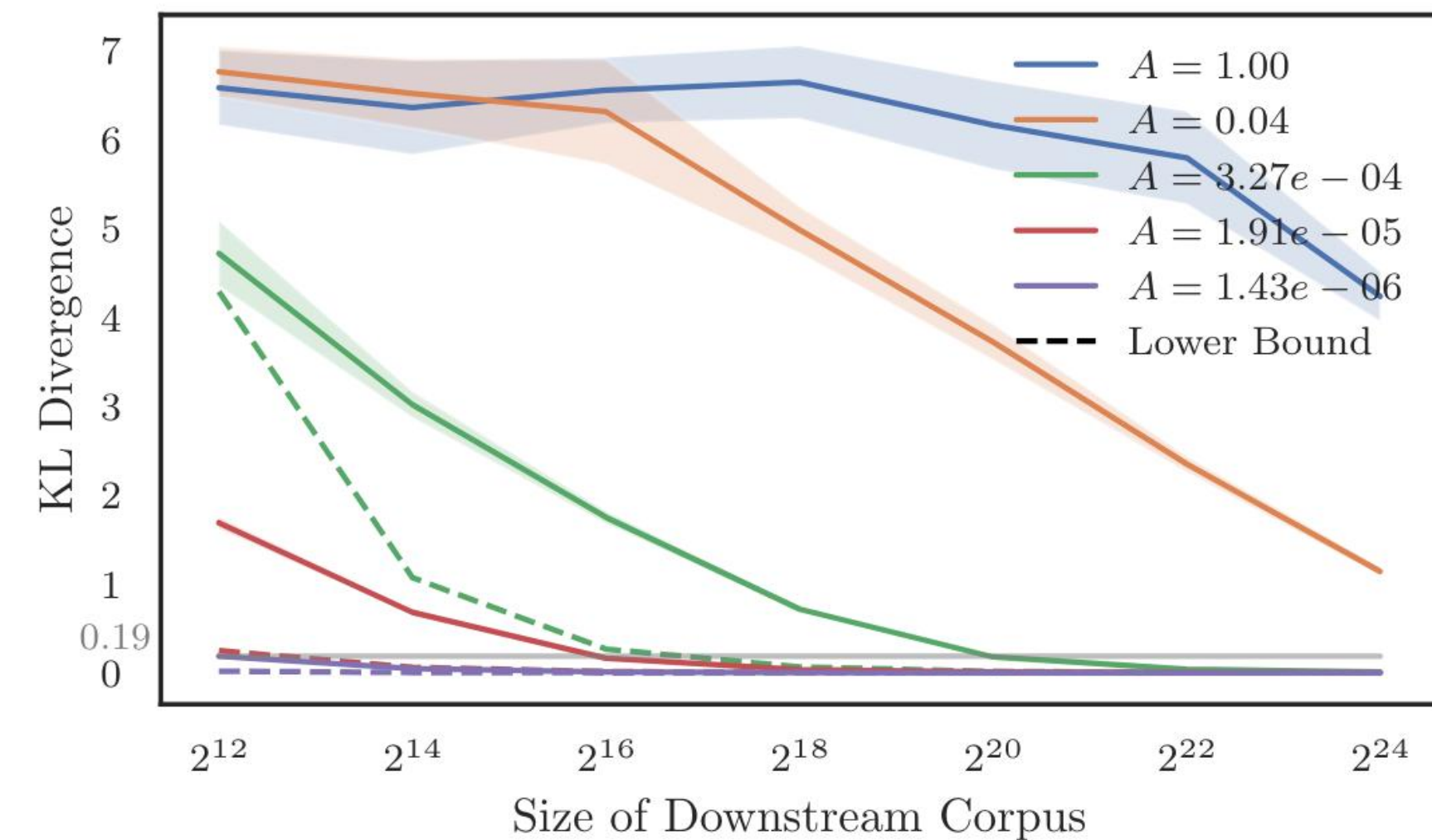


Figure 2: The KL divergence error in the first generation step on several synthetic relevance distribution data with different degrees of concentration. The vocabulary size is 2^{10} , the docID length is 3. A is the Simpson diversity index of the relevance distribution. As for highly concentrated distributions, e.g., $A = 1$, and $A = 0.04$, the Lyapunov's condition will no longer hold (see Theorem B.3 for more details), we do not draw their lower bounds.

$$\text{KL} [\text{Pr}(\cdot | C) \parallel \text{Pr}(\cdot | C_i)] \gtrsim \frac{0.05A}{p}. \quad (11)$$

» Experiments on real-world dataset

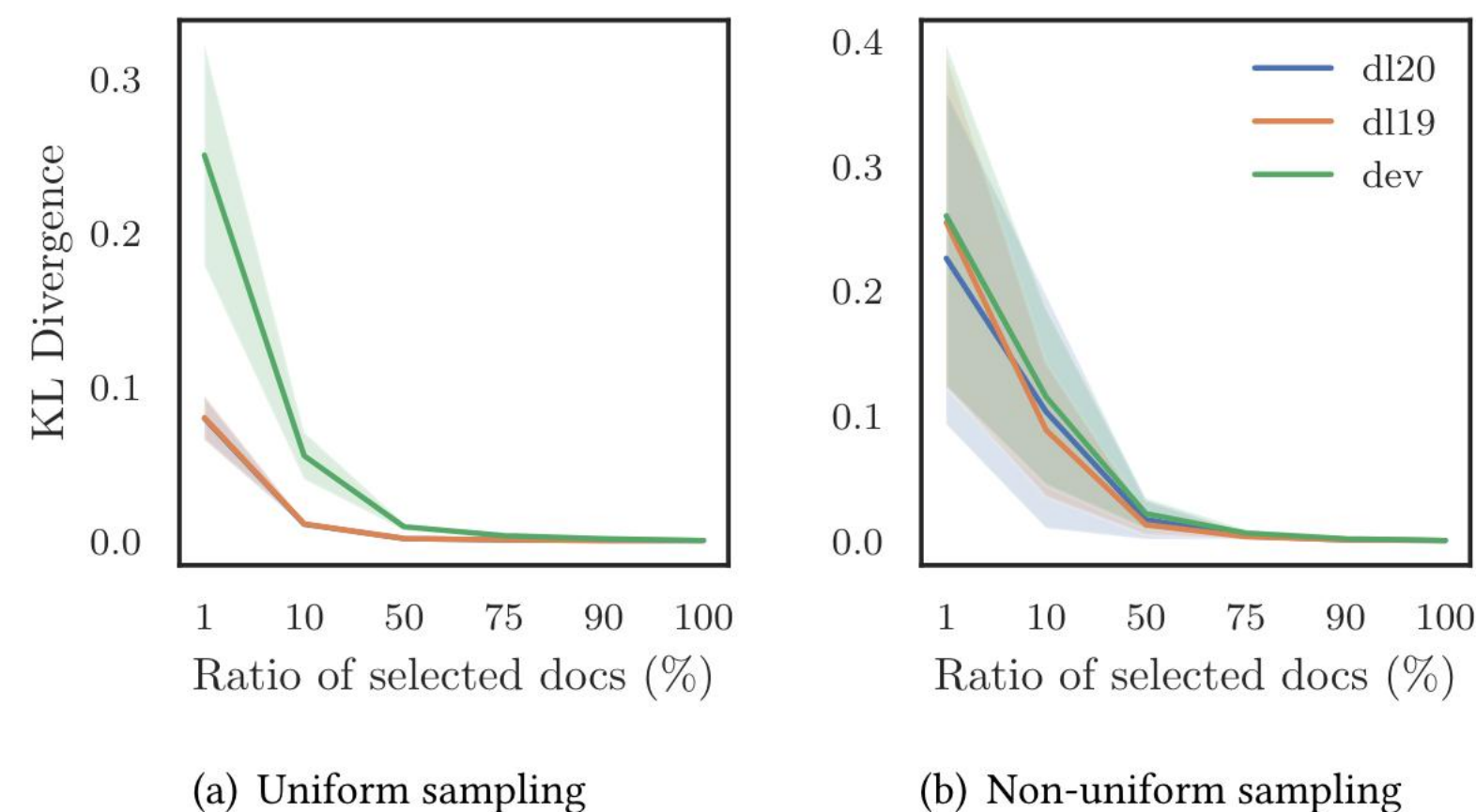


Figure 5: The KL divergence error in the first generation step on MS MARCO V1 passage corpus. A subset of top-10,000 rank list from SLIM++ [25] is treated as the complete corpus, and the downstream corpus is (a) uniformly sampled, or (b) sampled with the relevance distribution.

Table 2: Recall@50 and Precision@1 of relevant branches (256 in total) at the first generation step, which are denoted as “R@50” and “P@1”. The highest 50 documents are labeled relevant, and the branches contain these documents are labeled relevant. The relevance distribution is constructed in Section 7.1.

Model	R@50	P@1
TREC DL 19	53.7	69.8
TREC DL 20	63.3	75.9
MS MARCO-dev	67.5	90.5

- The error is substantially higher than in the uniform case.
- Non-uniform sampling introduces greater variance and exacerbates the mismatch between distributions.
- The recall is relatively low.
- The top-1 precision is higher.

» Limitations

- Fail to provide practical assumptions on the relevance distribution and the structure of docID.
- The results are sensitive to the parameters and assumptions and may not accurately reflect practical real-world situations.
- Have not studied how the two factors affect each other when using constrained beam search.
- The selection of docid is limited.

» Conclusion

- For the effect of **constraints**, we derive a lower bound of the error, which arises due to the **unawareness of future constraints** during generation and is shown to **depend on the average Simpson diversity index** of the relevance distribution.
- For the **beam search algorithm** used during generation, we prove that for sparse relevance distributions, beam search can achieve **perfect top-1 precision** but suffer from **poor top- k recall performance**.



感谢大家耐心倾听！